

Лабораторная работа №12

Настройка NAT

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Информация

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Выполнение работы

```
provider-gspolyakov-gw-1>enable
provider-gspolyakov-gw-1#confi
provider-gspolyakov-gw-1#configure ter
provider-gspolyakov-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
provider-gspolyakov-gw-1(config)#line vty 0 4
provider-gspolyakov-gw-1(config-line)#passw
provider-gspolyakov-gw-1(config-line)#password cisco
provider-gspolyakov-gw-1(config-line)#login
provider-gspolyakov-gw-1(config-line)#exit
provider-gspolyakov-gw-1(config)#line console 0
provider-gspolyakov-gw-1(config-line)#password cisco
provider-gspolyakov-gw-1(config-line)#login
provider-gspolyakov-gw-1(config-line)#exit
provider-gspolyakov-gw-1(config)#enable secret cisco
provider-gspolyakov-gw-1(config)#servi
provider-gspolyakov-gw-1(config)#service pa
provider-gspolyakov-gw-1(config)#service password-encryption
provider-gspolyakov-gw-1(config)#us
provider-gspolyakov-gw-1(config)#username admin pr
provider-gspolyakov-gw-1(config)#username admin privilege 1 se
provider-gspolyakov-gw-1(config)#username admin privilege 1 secret cisco
provider-gspolyakov-gw-1(config)#
```

```
provider-gspolyakov-sw-1(config-if)#exit
provider-gspolyakov-sw-1(config)#line vty 0 4
provider-gspolyakov-sw-1(config-line)#password cisco
provider-gspolyakov-sw-1(config-line)#login
provider-gspolyakov-sw-1(config-line)#exit
provider-gspolyakov-sw-1(config)#line conso
provider-gspolyakov-sw-1(config)#line console 0
provider-gspolyakov-sw-1(config-line)#pass
provider-gspolyakov-sw-1(config-line)#password cisco
provider-gspolyakov-sw-1(config-line)#login
provider-gspolyakov-sw-1(config-line)#exit
provider-gspolyakov-sw-1(config)#enabl
provider-gspolyakov-sw-1(config)#enable se
provider-gspolyakov-sw-1(config)#enable secret cisco
provider-gspolyakov-sw-1(config)#serv
provider-gspolyakov-sw-1(config)#service paa
provider-gspolyakov-sw-1(config)#service pas
provider-gspolyakov-sw-1(config)#service password-encryption
provider-gspolyakov-sw-1(config)#us
provider-gspolyakov-sw-1(config)#username admin pr
provider-gspolyakov-sw-1(config)#username admin privilege 1 se
provider-gspolyakov-sw-1(config)#username admin privilege 1 secret cisco
provider-gspolyakov-sw-1(config)#
```

```
provider-gspolyakov-gw-1(config-if)#exit
provider-gspolyakov-gw-1(config)#interface FastEthernet0/0
provider-gspolyakov-gw-1(config-if)#
provider-gspolyakov-gw-1(config-if)#exit
provider-gspolyakov-gw-1(config)#interface FastEthernet0/1
provider-gspolyakov-gw-1(config-if)#no ip address
provider-gspolyakov-gw-1(config-if)#ip address 192.0.2.1 255.255.255.0
provider-gspolyakov-gw-1(config-if)#ip address 192.0.2.1 255.255.255.0
provider-gspolyakov-gw-1(config-if)#exit
provider-gspolyakov-gw-1(config)#int
provider-gspolyakov-gw-1(config)#interface f0/0
provider-gspolyakov-gw-1(config-if)#no sh
provider-gspolyakov-gw-1(config-if)#no shutdown
provider-gspolyakov-gw-1(config-if)#exit
provider-gspolyakov-gw-1(config)#int
provider-gspolyakov-gw-1(config)#interface f0/0.4 enc
provider-gspolyakov-gw-1(config)#interface f0/0.4
provider-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.4, changed state to up
enc
provider-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 4
provider-gspolyakov-gw-1(config-subif)#ip address 198.51.100.1
% Incomplete command.
provider-gspolyakov-gw-1(config-subif)#ip address 198.51.100.1 255.255.255.0
% 198.51.100.0 overlaps with FastEthernet0/0
provider-gspolyakov-gw-1(config-subif)#
provider-gspolyakov-gw-1(config-subif)#ex
provider-gspolyakov-gw-1(config-subif)#exit
provider-gspolyakov-gw-1(config)#interface f0/0.4
provider-gspolyakov-gw-1(config-subif)#des
provider-gspolyakov-gw-1(config-subif)#description internet
provider-gspolyakov-gw-1(config-subif)#exit
provider-gspolyakov-gw-1(config)#exit
```

```
provider-gspolyakov-sw-1(config)#interface f0/1
provider-gspolyakov-sw-1(config-if)#sw
provider-gspolyakov-sw-1(config-if)#switchport m
provider-gspolyakov-sw-1(config-if)#switchport mode t
provider-gspolyakov-sw-1(config-if)#switchport mode trunk

provider-gspolyakov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ex
provider-gspolyakov-sw-1(config-if)#exit
provider-gspolyakov-sw-1(config)#interface f0/2
provider-gspolyakov-sw-1(config-if)#switchport mode trunk

provider-gspolyakov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
exit
provider-gspolyakov-sw-1(config)#
```



```
provider-gspolyakov-sw-1(config)#vlan 4
provider-gspolyakov-sw-1(config-vlan)#name nat
provider-gspolyakov-sw-1(config-vlan)#exit
provider-gspolyakov-sw-1(config)#inter
provider-gspolyakov-sw-1(config)#interface vlan4
provider-gspolyakov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan4, changed state to up
no xh
provider-gspolyakov-sw-1(config-if)#no xs
provider-gspolyakov-sw-1(config-if)#no sh
provider-gspolyakov-sw-1(config-if)#no shutdown
provider-gspolyakov-sw-1(config-if)#exit
provider-gspolyakov-sw-1(config)#
```

```
msk-donskaya-gspolyakov-gw-1(config)#interface f0/1
msk-donskaya-gspolyakov-gw-1(config-if)#no
msk-donskaya-gspolyakov-gw-1(config-if)#no sh
msk-donskaya-gspolyakov-gw-1(config-if)#no shutdown
msk-donskaya-gspolyakov-gw-1(config-if)#exit
msk-donskaya-gspolyakov-gw-1(config)#interface f0/1.4
msk-donskaya-gspolyakov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.4, changed state to up

msk-donskaya-gspolyakov-gw-1(config-subif)#enca
msk-donskaya-gspolyakov-gw-1(config-subif)#encapsulation dot1Q 4
msk-donskaya-gspolyakov-gw-1(config-subif)#ip add
msk-donskaya-gspolyakov-gw-1(config-subif)#ip address 198.51.100.2
% Incomplete command.
msk-donskaya-gspolyakov-gw-1(config-subif)#ip address 198.51.100.2 255.255.255.0
% 198.51.100.0 overlaps with FastEthernet0/1
msk-donskaya-gspolyakov-gw-1(config-subif)#ip address 198.51.100.2 255.255.255.240
% 198.51.100.0 overlaps with FastEthernet0/1
msk-donskaya-gspolyakov-gw-1(config-subif)#description internet
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#ip rou
msk-donskaya-gspolyakov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 198.51.100.2 198.51.100.14 netma
msk-donskaya-gspolyakov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 198.51.100.1
msk-donskaya-gspolyakov-gw-1(config)#exit
msk-donskaya-gspolyakov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

```
msk-donskaya-gspolyakov-gw-1(config)#ip nat pool main-pool 198.51.100.2 198.51.100.14 netmask 255.255.255.240  
msk-donskaya-gspolyakov-gw-1(config)#ip nat
```

```
msk-donskaya-gspolyakov-gw-1(config)#ip access-list extended nat-inet
```

```
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#remark dk
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#permit tcp 10.128.3.0 0.0.0.255 host 192.0.2.11 eq 80
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#permit tcp 10.128.3.0 0.0.0.255 host 192.0.2.12 eq 80
```

```
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#remark departaments  
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#permit tcp 10.128.4.0 0.0.0.255 host 192.0.2.13 eq 80
```

```
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#remark adm  
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#permit tcp 10.128.5.0 0.0.0.255 host 192.0.2.14 eq 80
```

```
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#remark admin  
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#permit tcp 10.128.6.0 0.0.0.255 host 192.0.2.14 eq 80
```



```
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#remark admin  
msk-donskaya-gspolyakov-gw-1(config-ext-nacl)#permit ip host 10.128.6.0 any
```

```
msk-donskaya-gspolyakov-gw-1(config)#ip nat inside source list nat-inet pool main-pool overload  
msk-donskaya-gspolyakov-gw-1(config) #
```

```
msk-donskaya-gspolyakov-gw-1(config)#int f0/0.3
msk-donskaya-gspolyakov-gw-1(config-subif)#ip nat inside
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#int f0/0.101
msk-donskaya-gspolyakov-gw-1(config-subif)#ip nat inside
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#int f0/0.102
msk-donskaya-gspolyakov-gw-1(config-subif)#ip nat inside
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#int f0/0.103
msk-donskaya-gspolyakov-gw-1(config-subif)#ip nat inside
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#int f0/0.104
msk-donskaya-gspolyakov-gw-1(config-subif)#ip nat inside
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
msk-donskaya-gspolyakov-gw-1(config)#int f0/1.4
msk-donskaya-gspolyakov-gw-1(config-subif)#ip nat ou
msk-donskaya-gspolyakov-gw-1(config-subif)#ip nat outside
msk-donskaya-gspolyakov-gw-1(config-subif)#exit
```

```
msk-donskaya-gspolyakov-gw-1(config)#ip nat inside source static tcp 10.128.0.2 80 198.51.100.2 80
```

```
msk-donskaya-gspolyakov-gw-1(config)#ip nat inside source static tcp 10.128.0.3 20 198.51.100.3 20
```

```
msk-donskaya-gspolyakov-gw-1(config)#ip nat inside source static tcp 10.128.0.3 21 198.51.100.3 21
```

```
msk-donskaya-gspolyakov-gw-1(config)#ip nat inside source static tcp 10.128.0.4 25 198.51.100.4 25  
msk-donskaya-gspolyakov-gw-1(config)#ip nat inside source static tcp 10.128.0.4 110 198.51.100.4 110
```

```
msk-donskaya-gspolyakov-gw-1(config)#ip nat inside source static tcp 10.128.6.200 3389 198.51.100.10 3389  
msk-donskaya-gspolyakov-gw-1(config)#
```